

## Part "B" - System Design Scenarios

Hotel Booking System.

### PAC - Problem Analysis Chart

#### Given Data

- Number of guests (N),
- season details
- Room type.
- Nights stayed.

#### Required Results

- Final price for each guest.
- Total hotel revenue.

#### Processing Required

- Take input from the user.
- Determine room rate based on season and room type.
- Calculate room cost.
- Apply 15% discount if nights stayed exceeds 7.
- Display the final cost
- Add each guest's price to total revenue
- Display total revenue at the end.

#### Solution Alternatives

- Use of nested else if statement.
- Use of switch-case for room type and season rates



Date:

# Hotel Booking System. IPO Chart

| Input                                                                                                                         | Process                                                                                                                                                                                                                                                                                                                                               | Output                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Number of guest N</li> <li>Season</li> <li>Room type</li> <li>Nights stayed</li> </ul> | <ul style="list-style-type: none"> <li>process input for each guest</li> <li>Determine applicable room rate</li> <li>Calculate room cost based on rate and nights</li> <li>Check whether long-stay discount applies</li> <li>Apply discount when applicable</li> <li>Calculate final price</li> <li>sum all guest prices for hotel revenue</li> </ul> | <ul style="list-style-type: none"> <li>Each guest's total price</li> <li>Hotel total revenue</li> </ul> |



# Algorithm

1. Start
2. Take  $N$  as Input from user
3. Set  $x \leftarrow 0$
4. set previous  $\leftarrow 0$
5. while  $N > x$  do the following
  - set discount = 0%
  - increment  $x$  by 1
  - Take input for season, room type, and nights stayed.
  - Switch (season)
    - Case 1:
      - Switch (rooms)
        - Case 1:
          - Set Rate = 8000
        - Case 2:
          - Set Rate = 8000
        - Case 3:
          - Set Rate = 12000
      - Case 2:
        - Switch (rooms)
          - Case 1:
            - Set Rate = 3000
          - Case 2:
            - Set Rate = 5000
          - Case 3:
            - Set Rate = 8000
    - calculate price by  $\text{rate} * \text{nights}$



Date:

- if nights  $> 7$  do the following  
discount = price  $\times$  0.15
- total\_price = (rate  $\times$  nights) - discount
- Display total\_price
- Now add total-price and previous and assign it in total-price
- assign total-price value to previous.

6. Display total hotel revenue by printing total\_price
7. stop



## Pseudocode

Begin

Declare  $N, x, \text{discount}, \text{previous}, \text{nights}, \text{rate}, \text{season}$   
 $\text{room\_type}, \text{total\_price}, \text{price}$  as integers.

Input  $N$

let  $x \leftarrow 0$

let  $\text{previous} \leftarrow 0$

while  $N > x$

let  $\text{discount} \leftarrow 0$

$x \leftarrow x + 1$

Input  $\text{season}$  ( $\text{peak} = 1, \text{off-peak} = 2$ )

Input  $\text{room}$  ( $\text{standard} = 1, \text{Deluxe} = 2, \text{suite} = 3$ )

Input  $\text{nights\_stayed}$

switch ( $\text{season}$ )

Case 1:

• switch ( $\text{rooms}$ )

Case 1:

{ let  $\text{Rate} \leftarrow 5000$  }

Case 2:

{ let  $\text{Rate} \leftarrow 8000$  }

Case 3:

{ let  $\text{Rate} \leftarrow 12000$  }

Case 2:

Switch ( $\text{rooms}$ )

Case 1

{ let  $\text{Rate} \leftarrow 3000$  }

Case 2

{ let  $\text{Rate} \leftarrow 5000$  }

Case 3

{ let  $\text{Rate} \leftarrow 8000$  }



Date:

•  $\text{price} \leftarrow \text{rate} * \text{night}$

if  $\text{night} > 7$

$\text{discount} \leftarrow \text{price} * 0.15$

end if

$\text{total\_price} = \text{price} - \text{discount}$

Output  $\text{total\_price}$  (guest total)

$\text{total\_price} \leftarrow \text{total\_price} + \text{previous}$

$\text{previous} \leftarrow \text{total\_price}$

• end while

• Output  $\text{total\_price}$  (revenue)

• End



# Flowchart



